

THEORIES OF FAILURE

Introduction

- In case of material subjected to simple state of stress (tension or compression), failure occurs when the stress in the material reaches the elastic limit stress.
- In case of material subjected to complex stresses, the stage of failure is determined either to practically or theoretically.
- Since the complex stress system can be simplified into three principal stresses, the problem reduced to linking the three principal stresses to the stresses at elastic limit in case of simple stresses.

The common most theories are

1. Maximum principal stress theory
2. Maximum Principal strain theory
3. Maximum shear stress theory
4. Maximum strain energy theory
5. Maximum shear strain energy theory

Maximum principal stress theory

- This theory was proposed by Rankine.
- It states that failure will occur when the maximum principal stress (σ_1) in the complex system reaches the value of maximum stress (σ_{yt}) at the elastic limit in simple tension or the minimum principal stress (i.e. maximum principal compression stress) reaches the elastic limit (σ_{yc}) in simple compression.

$$\sigma_1 = \sigma_{yt} \quad \text{in simple tension}$$

$$|\sigma_3| = \sigma_{yc} \quad \text{in simple compression}$$

- For the design, the maximum principal stress should not exceed the working stress σ for the material. $\sigma_1 \leq \sigma$,

- Working stress, $\sigma = \frac{\sigma_y}{F}$

F : Factor of safety

- This theory is valid for brittle metals such as cast iron.

Maximum Principal strain theory

- This theory was proposed by Saint Venant.
- It states that the failure of a material occurs when the major principal tensile strain reaches the strain at the elastic limit in simple tension or when the minor principal strain (i.e maximum principal compressive strain) reaches the strain at elastic limit in simple compression.
- This theory is more appropriate for ductile materials, brittle materials and materials under hydrostatic pressure.

Maximum Principal strain theory

Principal strain in the direction of principal stress σ_1 , $e_1 = \frac{1}{E}[\sigma_1 - \mu(\sigma_2 + \sigma_3)]$

Principal strain in the direction of principal stress σ_3 , $e_3 = \frac{1}{E}[\sigma_3 - \mu(\sigma_1 + \sigma_2)]$

According to maximum principal strain theory, the conditions to cause failure are

$$e_1 > \frac{\sigma_{yt}}{E} ; \quad \frac{1}{E}[\sigma_1 - \mu(\sigma_2 + \sigma_3)] > \frac{\sigma_{yt}}{E} \Rightarrow \sigma_1 - \mu(\sigma_2 + \sigma_3) > \sigma_{yt} \quad \text{or}$$

$$|e_3| > \frac{\sigma_{yc}}{E} ; \quad \frac{1}{E}[\sigma_3 - \mu(\sigma_1 + \sigma_2)] > \frac{\sigma_{yc}}{E} \Rightarrow \sigma_3 - \mu(\sigma_1 + \sigma_2) > \sigma_{yc}$$

To prevent failure

$$\sigma_1 - \mu(\sigma_2 + \sigma_3) < \sigma_{yt}$$

$$\sigma_3 - \mu(\sigma_1 + \sigma_2) < \sigma_{yc}$$

Maximum shear stress theory

- This theory is also called Coulomb Guest's or Treasca's theory
- It states that the material will fail when the maximum shear stress ($\tau_{\max}$) in the complex system reaches the value of maximum shear stress in simple tension at the elastic limit.

$$\tau_{\max} = \frac{\sigma_1 - \sigma_3}{2} = \frac{\sigma_{yt}}{2}$$

$$\sigma_1 - \sigma_3 = \sigma_{yt}$$

$$\sigma_1 - \sigma_3 = \sigma_t$$

Maximum strain energy theory

- This theory was proposed by Beltrani-Haigh.
- It states that the failure of a material occurs when the total strain energy in the material reaches the total strain energy of the material at the elastic limit in simple tension.

In a 3D stress system, the strain energy per unit volume is given by

$$U = \frac{1}{2E} \left[\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\mu(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1) \right]$$

At the point of failure

$$\frac{1}{2E} \left[\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\mu(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1) \right] = \frac{\sigma_y^2}{2E}$$

$$\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\mu(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1) = \sigma_y^2$$

In 2D stress system ($\sigma_2 = 0$), the above equation reduce to

$$\sigma_1^2 + \sigma_3^2 - 2\mu\sigma_1\sigma_3 = \sigma_y^2$$

For the design

$$\sigma_1^2 + \sigma_3^2 - 2\mu\sigma_1\sigma_3 \leq \sigma^2$$

Maximum shear strain energy theory

Distortion energy theory

- This theory was proposed by Von Mises-Henky
- It states that the elastic failure occurs when the shear strain energy per unit volume in the stressed material reaches a value equal to the shear strain energy per unit volume at the elastic limit point in tension.

Shear strain energy per unit volume due to principal stresses σ_1, σ_2 and σ_3 ,

$$U_s = \frac{1+\mu}{3E} [\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - (\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1)]$$

$$U_s = \frac{1+\mu}{6E} [(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]$$

$$U_s = \frac{1}{12G} [(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]$$

Maximum shear strain energy theory

Distortion energy theory

For the simple tension at the elastic limit point, $(\sigma_1 = \sigma_{yt}, \sigma_2 = \sigma_3 = 0)$, the shear strain energy per unit volume is given by

$$U'_s = \frac{1}{12G} [(\sigma_{yt} - 0)^2 + (0 - 0)^2 + (0 - \sigma_{yt})^2] = \frac{1}{12G} \cdot 2\sigma_{yt}^2$$

Equating the two strain energies, $(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 = 2\sigma_t^2$

In 2D stress system ($\sigma_2 = 0$), the above equation reduce to

$$\sigma_1^2 + \sigma_3^2 - \sigma_1\sigma_3 = \sigma_t^2$$

Design conditions for various failure theory

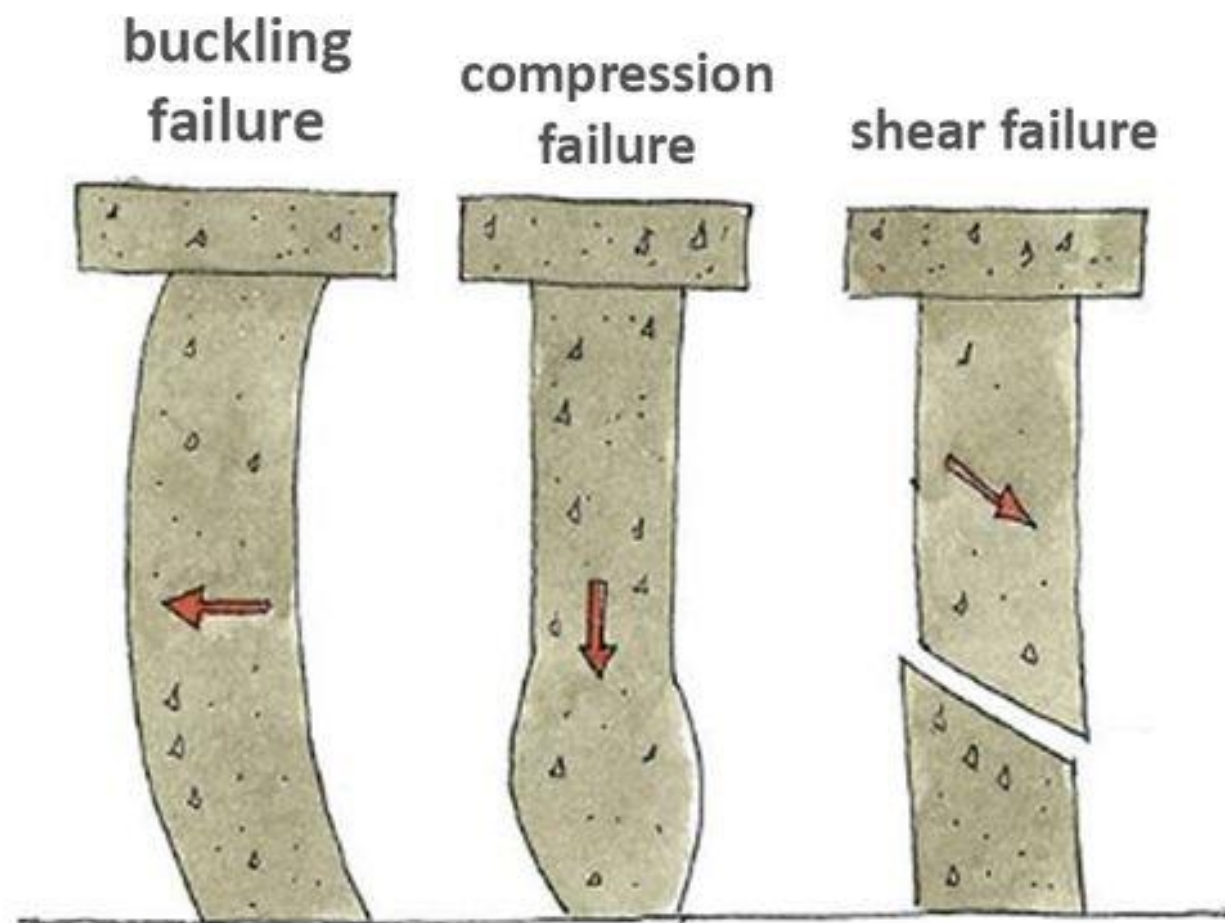
Failure theory	Proposed by	Condition for design
Maximum principal stress theory	Rankine, Lamé	$\sigma_1 \leq \sigma$
Maximum principal strain theory	Saint Venants	$\frac{1}{E}(\sigma_1 - \mu\sigma_2) \leq \frac{\sigma_y}{E}$
Maximum shear stress theory	Coulomb Guest, Tresca	$\frac{\sigma_1 - \sigma_2}{2} \leq \frac{\sigma_y}{2}$
Maximum strain energy theory	Beltrami-Haigh	$(\sigma_1^2 + \sigma_2^2 - 2\mu\sigma_1\sigma_2) \leq \sigma_y^2$
Distortion energy theory	Huber-Henky-Von Mises	$(\sigma_1^2 + \sigma_2^2 - \sigma_1\sigma_2) \leq \sigma_y^2$

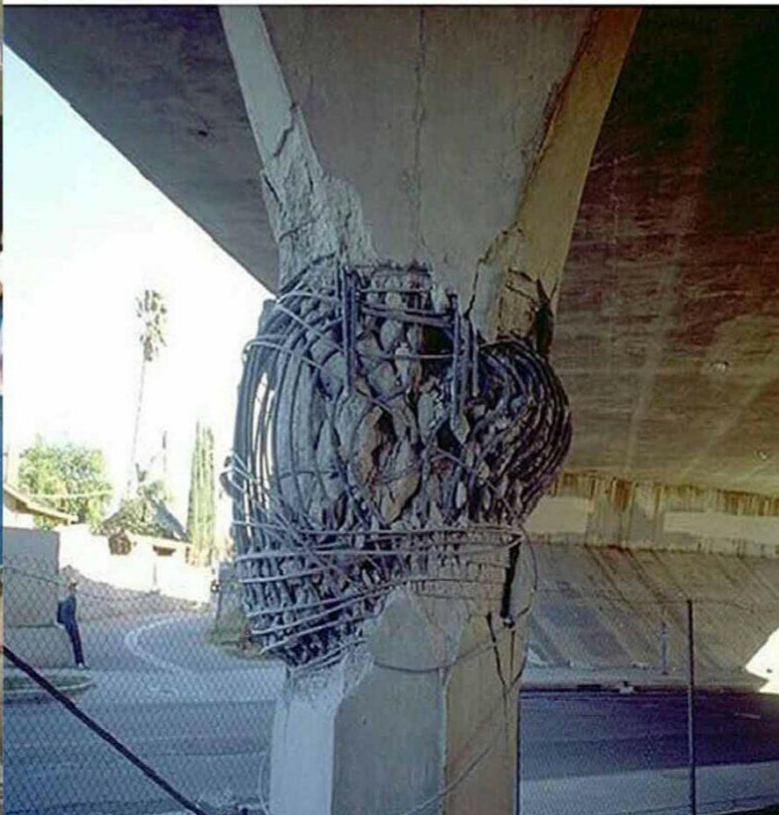
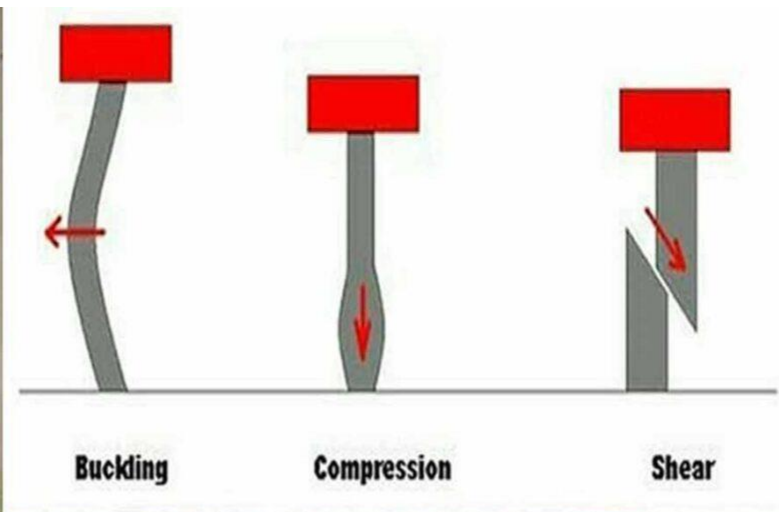
Theories applicable Brittle and Ductile materials

Theories Applicable to Ductile Materials	Theories Applicable to Brittle Materials
Maximum Shear Stress Theory (Tresca) Ductile materials fail due to shear, making this highly applicable	Maximum Principal Stress Theory Brittle materials typically fail when the normal stress exceeds their tensile or compressive strength
Maximum Shear Strain Energy Theory (Von Mises) A more accurate theory for ductile materials under complex stress states	Maximum Principal Strain Theory Occasionally used for brittle materials, as strain effects are significant during fracture
Maximum Strain Energy Theory Can be used for ductile materials but is less common than von Mises	

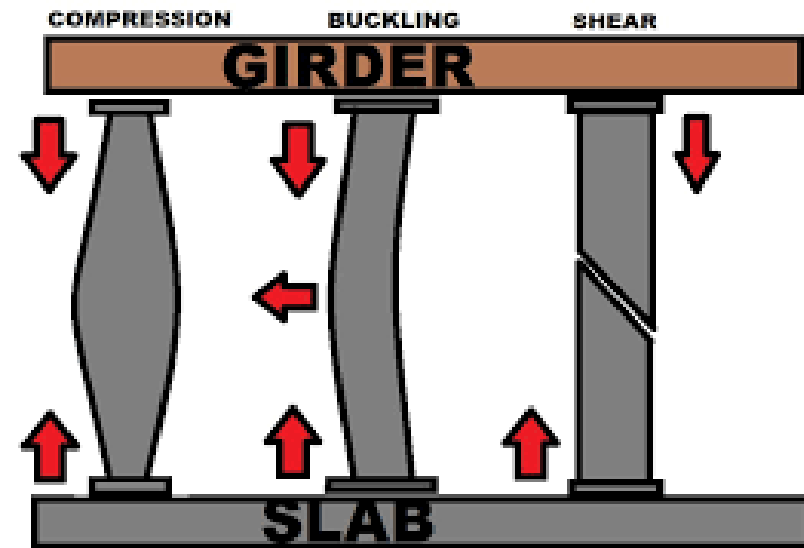
Columns and Struts

Types of Column Failure





TYPES OF COLUMN FAILURE

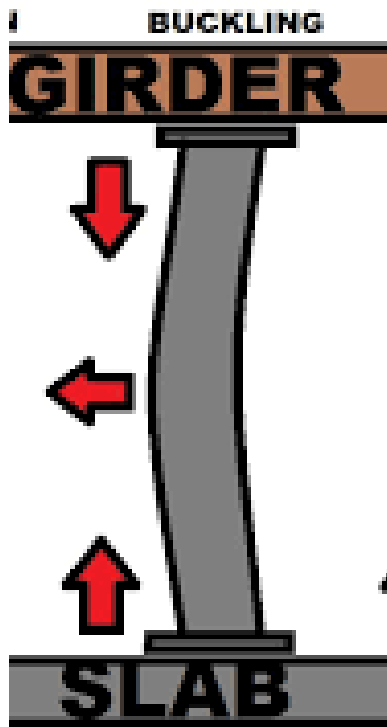


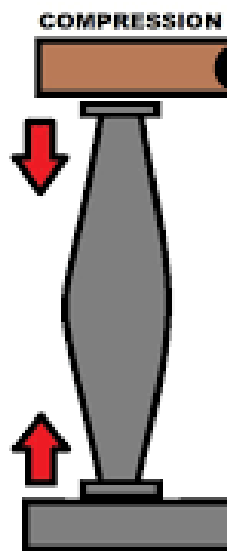


SHEAR
FAILURE

SHEAR
FAILURE

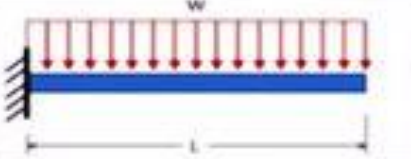
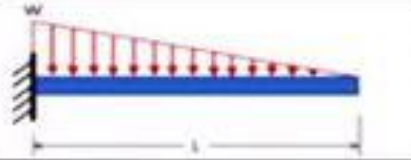
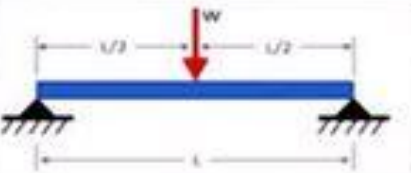
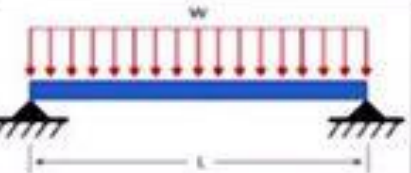






Compression Failure

S.No.	End conditions of column	Crippling load in terms of		Relation between effective length and actual length
		Actual length	Effective length	
1.	Both ends hinged	$\frac{\pi^2 EI}{l^2}$	$\frac{\pi^2 EI}{L_e^2}$	$L_e = l$
2.	One end is fixed and other is free	$\frac{\pi^2 EI}{4l^2}$	$\frac{\pi^2 EI}{L_e^2}$	$L_e = 2l$
3.	Both ends fixed	$\frac{4\pi^2 EI}{l^2}$	$\frac{\pi^2 EI}{L_e^2}$	$L_e = \frac{l}{2}$
4.	One end fixed and other is hinged	$\frac{4\pi^2 EI}{l^2}$	$\frac{\pi^2 EI}{L_e^2}$	$L_e = \frac{l}{\sqrt{2}}$

SR. NO.	TYPE OF BEAM	MAX. BM	SLOPE	DEFLECTION
1		M	$\theta = \frac{ML}{EI} = \frac{ML}{EI}$	$\delta = \theta \times \frac{L}{2} = \frac{ML^2}{2EI}$
2		WL	$\theta = \frac{ML}{2EI} = \frac{WL^2}{2EI}$	$\delta = \theta \times \frac{2L}{3} = \frac{WL^3}{3EI}$
3		$\frac{WL^2}{2}$	$\theta = \frac{ML}{3EI} = \frac{WL^3}{6EI}$	$\delta = \theta \times \frac{3L}{4} = \frac{WL^4}{8EI}$
4		$\frac{WL^2}{6}$	$\theta = \frac{ML}{4EI} = \frac{WL^3}{24EI}$	$\delta = \theta \times \frac{4L}{5} = \frac{WL^4}{30EI}$
5		$\frac{WL}{4}$	$\theta = \frac{ML}{4EI} = \frac{WL^2}{16EI}$	$\delta = \theta \times \frac{L}{3} = \frac{WL^3}{48EI}$
6		$\frac{WL^2}{8}$	$\theta = \frac{ML}{3EI} = \frac{WL^3}{24EI}$	$\delta = \theta \times \frac{5L}{16} = \frac{5WL^4}{384EI}$